



Muhammed Shafeeh K

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PROFESSIONAL SUMMARY

Accomplished machine learning engineer with a strong history of creating cutting-edge solutions for complex challenges. Dedicated to applying advanced machine learning techniques to achieve significant business outcomes and measurable results. Enthusiastic about bringing my expertise and leadership to a dynamic role within a global organization.

EDUCATION

DEGREE | Mechanical Engineering | College of Engineering Trivandrum (APJ KTU), India 2016– 2020

TECH STACK

Machine Learning Algorithms	Linear Regression, Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, K-Nearest Neighbors, Neural Networks, And Deep Learning Architectures. Computer Vision, Data Preprocessing, Feature Engineering And Model Evaluation NLP, CNN, RNN, LSTM, LLM, Transformers, GenAI
Deep Learning Frameworks	TensorFlow, PyTorch, sci-kit-learn, Keras, Open CV
Data Processing and Analysis	NumPy, Pandas, SciPy, SQL, NoSQL, Tableau.
Model Evaluation and Validation	Cross-validation, Hyperparameter Tuning, Model Selection, MLOPs
Programming Languages:	Python, C++
IDE	Anaconda, Jupyter Notebook, GitHub Copilot, VS Code
Cloud Computing Platforms	AWS (AWS S3, Amazon SageMaker, Amazon EC2, AWS Lambda, Amazon ECR, Amazon ECS)
Deployment and Productionisation	Docker, Kubernetes

Natural Language Processing (NLP)	NLTK, spaCy, Transformers, GenAI, RAG, Vector DB, Langchain, Huggingface, Prompt Engineering
Computer Vision	OpenCV, TensorFlow Object Detection API, Dlib
Project Management	Agile, Scrum, Jira

PROFESSIONAL EXPERIENCE

MACHINE LEARNING ENGINEER | QZIGMA TECHNOLOGIES | Kerala, India Jan 2025 – Present

- Developing and implementing machine learning algorithms and models.
- Data pre-processing, feature engineering, and data cleaning.
- Collaborating with cross-functional teams to gather requirements and deploy machine learning solutions.
- Conducting exploratory data analysis (EDA) to gain insights from data.
- Optimizing models for performance, scalability, and efficiency.
- Evaluating and selecting appropriate machine learning techniques and algorithms.

MACHINE LEARNING ENGINEER | RANMARINE | Rotterdam, Netherlands Jan 2021 – Oct 2024

- Developed and implemented machine learning algorithms for various applications, including natural language processing, computer vision, and predictive analytics.
- Conducted data preprocessing, feature engineering, and model training using Python, TensorFlow, and Scikit-learn.
- Collaborated with cross-functional teams to integrate machine learning models into production systems and optimize performance.
- Contributed to research and development projects, staying abreast of the latest advancements in machine learning technologies.
- Developed and implemented deep learning frameworks like TensorFlow, PyTorch, and Keras.
- Involved in designing, implementing, and optimizing deep neural networks for tasks such as image recognition, natural language processing, and time series analysis.
- Conducted exploratory data analysis using libraries like NumPy, Pandas, and SciPy.
- Familiar with techniques such as normalization, scaling, dimensionality reduction, and outlier detection.
- Conducted evaluation and validation of machine learning models using appropriate metrics and techniques. And perform cross-validation, hyperparameter tuning, and model selection to ensure robust and generalizable performance.

MAJOR PROJECTS

Project: Medical Chatbot with Retrieval-Augmented Generation (RAG) (GenAI/NLP)

Description: Developed an AI-powered medical chatbot using Retrieval-Augmented Generation (RAG) to provide

accurate and context-aware responses for healthcare professionals and patients. Designed to handle a limited number of high-value medical queries, the chatbot assists with medical literature retrieval, symptom analysis, and guideline explanations while reducing misinformation.

Responsibilities:

- **LLM Integration:** Integrated large language models (GPT, BERT-based models) with a vector database (FAISS, Pinecone) to enhance retrieval-based responses.
- **Medical Data Processing:** Preprocessed medical literature and clinical guidelines using NLP techniques such as TF-IDF, and word embeddings.
- **Backend Development:** Built a FastAPI-based system to handle user queries, retrieve relevant context from medical sources, and generate responses.
- **Response Optimization:** Applied prompt engineering and ranking models to improve medical response accuracy and relevance.
- **Deployment & Compliance:** Deployed the chatbot on AWS (SageMaker, Lambda, EC2) with security and compliance measures aligned with healthcare standards.

Technologies Used:

- **Programming & AI Frameworks:** Python, LangChain, Hugging Face Transformers
- **Database & Retrieval:** FAISS, Pinecone
- **Web & Backend:** FastAPI, Docker
- **Cloud Deployment:** AWS (SageMaker, Lambda, EC2)

Outcomes:

- Increased accuracy of medical responses by 35% through improved document retrieval and ranking.
- Reduced misinformation risk by 40% using advanced validation techniques.
- Handled critical medical inquiries with a response time of <300ms while ensuring compliance with industry regulations.

Project: Floating Waste Detection & Autonomous Cleanup System (DL)

Description: Led the development of a computer vision system for an autonomous marine cleanup vessel to detect and classify floating waste (plastic, debris) in real time. The system was optimized for edge devices to ensure fast decision-making for navigation and waste collection in changing marine environments.

Responsibilities:

- **Initial Classification Prototyping:** Built a basic image classification model using VGG16 with transfer learning, achieving 89% accuracy on a dataset of marine waste images. This step helped me understand the features before moving to object detection.
- **Object Detection Model Development:** Used YOLOv5 for real-time waste detection, adjusting key settings like anchor boxes, learning rate, and IoU threshold to improve accuracy.
- **Dataset Preparation & Augmentation:** Collected and labeled a dataset of 20,000 marine images using Label Studio. Applied techniques like random cropping, blur, and color adjustments to improve model

performance in different lighting and weather conditions.

- **Image Preprocessing:** Improved image quality using OpenCV techniques like contrast adjustment (CLAHE) and multi-scale image processing to handle reflections on water.
- **Edge Deployment Optimization:** Converted the YOLOv5 model to TensorFlow Lite and ONNX runtime, achieving an 18ms/frame speed on NVIDIA Jetson Nano while maintaining 90% accuracy.

Technologies Used:

- **Machine Learning Frameworks:** TensorFlow, Keras, PyTorch
- **Object Detection:** YOLOv5 with tuning
- **Image Classification (Prototyping):** VGG16 with transfer learning
- **Image Processing:** OpenCV (contrast adjustment, multi-scale processing)
- **Data Annotation:** Label Studio
- **Version Control & Deployment:** GitHub Actions, Git
- **Edge & Cloud Computing:** NVIDIA Jetson.

Outcomes:

- Reduced false positives by 30% in challenging marine conditions (waves, glare) compared to baseline models.
- Improved waste collection efficiency by 15% through real-time detection-guided navigation.
- Deployed across 5 marine vessels, processing 1.2M+ frames daily with <5% CPU utilization on edge hardware.

Project: Fraud Detection System for E-commerce Transactions (ML)

Description: Designed and implemented a machine learning system to detect fraudulent e-commerce transactions in real-time, reducing financial losses and improving customer trust.

Responsibilities:

- Processed and analyzed 1 million+ transaction records, including user behavior, payment details, and device fingerprints, addressing class imbalance with SMOTE and undersampling techniques.
- Engineered 25+ behavioral and transactional features, such as average session duration, payment frequency, and device mismatch events, to enhance model performance.
- Trained and tested multiple classification models (Logistic Regression, Random Forest, XGBoost, LightGBM), achieving an F1-score of 91% with Random Forest after hyperparameter tuning using Bayesian Optimization.
- Implemented MLOps best practices, including CI/CD pipelines with GitHub Actions and model versioning using MLflow, ensuring reproducibility and streamlined model updates.
- Deployed the model as a microservice using FastAPI, ensuring integration with existing e-commerce platforms for real-time fraud detection.
- Integrated explainability techniques (e.g., SHAP and LIME) to provide insights into flagged transactions, such as unusual location-IP discrepancies or atypical purchase patterns.

Technologies: Python, sci-kit-learn, LightGBM, Pandas, FastAPI, SHAP, Docker, Kubernetes, AWS, Git.

Key Metrics:

- Precision: 89% | Recall: 93% | F1-score: 91%.
- Reduced manual fraud review workload by 40% while minimizing false positives.

Outcomes:

- Improved customer satisfaction by reducing delays in order processing caused by fraud checks.
- Presented the solution at an e-commerce innovation summit, earning recognition for the innovative use of ML in fraud prevention.

CERTIFICATIONS

- **Programming for Everybody (Getting Started with Python) | Coursera**
- **Foundations: Data, Data, Everywhere | Google | Coursera**
- **C for Everyone: Programming Fundamentals | UCSC | Coursera**
- **Machine Learning Specialization | Stanford | Coursera**